

Introduction to Artificial Intelligence

What is Artificial Intelligence?

Artificial Intelligence (AI) is the field of computer science concerned with building systems that can perform tasks that normally require human-like capabilities, such as perception, reasoning, learning, language understanding, and decision making.

Major Areas of AI

Important areas include machine learning, natural language processing, computer vision, robotics, knowledge representation, planning, and expert systems. Modern applications often combine several of these areas.

Machine Learning

Machine learning is a subset of AI in which a model learns patterns from data instead of being programmed with a separate rule for every possible case. Common learning settings are supervised learning, unsupervised learning, and reinforcement learning.

Applications

AI is used in recommendation systems, fraud detection, speech assistants, medical image analysis, route planning, search engines, industrial automation, and document question answering.

Limitations

AI systems can produce incorrect outputs, inherit patterns from training data, struggle with unfamiliar situations, and require careful evaluation. Good system design includes testing, monitoring, and human oversight where appropriate.

Key Terms

Model: a learned computational representation. Dataset: a collection of examples. Training: the process of fitting a model to data. Inference: using a trained model to produce an output for new input.